ASSIGNMENT-19.2

Task 1: Python ➜ Java (File Handling)

PYTHON CODE:-

```python
import os

filename = "input.txt"

# Create input.txt with default content if it does not exist
if not os.path.exists(filename):
    with open(filename, "w") as f:
        f.write("hello world example")

with open(filename, "r") as f:
    data = f.read()

with open("output.txt", "w") as f:
    f.write(data.upper())

print("Done")
```

```
Done
```

PROMPT:-

Convert this Python file handling code into Java with proper exception handling.

JAVA CODE:-

```python
import os

filename = "input.txt"

# Create input.txt with default content if it does not exist
if not os.path.exists(filename):
    with open(filename, "w") as f:
        f.write("hello world example")

with open(filename, "r") as f:
    data = f.read()

with open("output.txt", "w") as f:
    f.write(data.upper())

print("Done")
```

```
Done
```

Task 2: C++ ➜ Python (Sorting)

C++ CODE:-

```python
arr = [5, 3, 1, 4, 2]
n = len(arr)

# Bubble sort algorithm
for i in range(n):
    for j in range(0, n - i - 1):
        if arr[j] > arr[j + 1]:
            arr[j], arr[j + 1] = arr[j + 1], arr[j]  # Swap elements

# Print the sorted array
for x in arr:
    print(x, end=" ")
```

```
1 2 3 4 5
```

PROMPT:-

Convert the following C++ sorting program into Python.

PYTHON CODE:-

```python
arr = [5, 3, 1, 4, 2]

for i in range(len(arr)):
    for j in range(len(arr)-i-1):
        if arr[j] > arr[j+1]:
            arr[j], arr[j+1] = arr[j+1], arr[j]

print(arr)
```

```
[1, 2, 3, 4, 5]
```

Task 3: SQL ➜ Pandas

SQL QUERY:-

```sql
SELECT department, SUM(sales)
FROM data
GROUP BY department;
```

PROMPT:-

Convert the following SQL query into an equivalent Pandas DataFrame operation.

PANDAS CODE:-

```python
import pandas as pd
import os

# Create a sample data.csv file if it doesn't exist
filename = "data.csv"
if not os.path.exists(filename):
    with open(filename, "w") as f:
        f.write("department,sales\n")
        f.write("A,100\n")
        f.write("B,150\n")
        f.write("A,50\n")
        f.write("C,200\n")

df = pd.read_csv(filename)
result = df.groupby("department")["sales"].sum()

print(result)
```

```
department
A    150
B    150
C    200
Name: sales, dtype: int64
```

PYTHON(FLASK) CODE:-

```python
from flask import Flask
app = Flask(__name__)

@app.route('/')
def home():
    return "Hello"

app.run()
```

```
 * Serving Flask app '__main__'
 * Debug mode: off
INFO:werkzeug:WARNING: This is a development
 * Running on http://127.0.0.1:5000
INFO:werkzeug:Press CTRL+C to quit
```

PROMPT:-

Convert the following code from PYTHON FLASK to JAVA SPRING BOOT

JAVA(SPRING BOOT):-

```python
from flask import Flask
app = Flask(__name__)

@app.route('/')
def home():
    return "Hello"
```

TASK 5 – Python ➜ C++ (Even/Odd)

PYTHON CODE:-

```python
def check(n):
    if n % 2 == 0:
        print("Even")
    else:
        print("Odd")

check(5)
```

```
Odd
```

PROMPT:-

Convert the following Python function into a C++ program.

C++ CODE:-

```python
def check(n):
    if n % 2 == 0:
        print("Even")
    else:
        print("Odd")

# Example usage:
check(5) # This will print
# You can change the number
```

```
Odd
```